INCH-POUND

MIL-B-2427G (AR)
AMENDMENT 5
18 March 1991
SUPERSEDING
AMENDMENT 4
14 July 1989

MILITARY SPECIFICATION

BOX, AMMUNITION PACKING: WOOD, NAILED

This Amendment forms a part of Military Specification MIL-B-2427G (AR), dated 30 August 1984, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

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2.1.1 Specification and Standards: Under STANDARDS, MILITARY, delete:

"MS 35492 - Screw, Wood, Flat Heat, Cross Recessed
MS 35494 - Screw, Wood, Flat Head"

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3.3 Rope handles. After the third sentence, add:

"This length is the length of exposed rope. It does not include the length of rope needed for attachment to the box (i.e., that length attached to the cleats). This adds approximately six inches of rope."

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3.4 Metal fittings. Delete the following drawings listed:

"MS 35492 - Screw, Wood, Flat Head, Cross Recessed, Part Number (No.) MS-35492-41
MS 35494 - Screw, Wood, Flat Head, Part Number (No.) MS-35494-46"

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* Last sentence: Delete in its entirety and substitute:

"Screws shall be standard commercial number 7x3/4 inch or number 8x3/4 inch tamperproof wood or sheet metal screws with countersink heads. The screws shall be zinc or cadmium plated, and meet all requirements of FF-S-107 except for head design. Any style of tamperproof head is acceptable so long as the screws cannot be removed with a common hand tool."

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3.7.1 Tenth sentence: Delete "more than 1/8 the thickness of the board" and substitute "more than .125 inches below board surface".

3.7.1 After last sentence, add sentence as follows:

"When two nails are occasionally driven coincidentally in the same location, the nails shall be acceptable as long as the integrity of the box is not affected (splitting terminating in edge of box)."

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3.8 Box identification. After the third sentence, add:

"The letters "PC" shall be annotated on all boxes subjected to either the Cunapsol 5 (copper naphthenate) or the M-GARD W510 (copper naphthenate) preservative in accordance with 3.9."

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* 3.9 Preservative treatment. Delete in its entirety and substitute:

"3.9 Preservative treatment. Grade A boxes, individual or palletized or the finished wood parts thereof, shall be completely immersed for a minimum of one minute in a solution of wood preservative:

a. PQ56 reduced with water down to 1.8 percent copper-8-quinolinolate as solution (see 6.6).

b. An emulsion of wood preservative M-GARD W550 (Zinc naphthenate) reduced with water down to 3 percent zinc as metal (see 6.7).

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c. An emulsion of M-GARD W51Ø (copper naphthenate) reduced down with water to 2 percent copper as metal (see 6.14).

d. A solution of Cunapsol 5 (copper naphthenate) reduced down with water to 2 percent copper as metal (see 6.15).

Alternatively, Grade A boxes, individual or palletized or the finished wood parts thereof, shall be completely flooded for a minimum of one minute in PQ56, M-GARD W55Ø, M-GARD W51Ø or Cunapsol 5 preservative, as applicable. All interior and exterior surfaces (when finished wood parts are dipped) are to be completely inundated with preservative. Care shall be exercised to assure complete coverage of all surfaces of the board. Immediately following the dip treatment, wood products being treated shall be drained for a period of not less than 5 minutes. If wood products are palletized, the pallet load shall be tipped on edge to facilitate thorough drainage. After the dip treatment, the boxes must be air dried for a period of 24 hours minimum providing thru ventilation thus allowing full air circulation around all surfaces of the wood box. Accelerated drying in an oven or kiln is permitted providing oven or kiln temperature does not exceed 16Ø degrees F. The boxes must be air dried to a maximum moisture content of 18% when tested in accordance with 4.5.2 prior to shipment.

The box manufacturer will be required to obtain and provide all available safety, health and environmental data; i.e., EPA Hazard Data Sheets, OSHA Safety Data Sheets, etc. Of specific interest are the acute, subchronic and chronic toxicity data. Also, the manufacturer will obtain and provide any special safety, health and environmental information (apparatus and procedures) to be used throughout the treated box duty life and disposal.

3.9.1 Presence of PQ56 (copper-8-quinolinolate) preservative. When treated with PQ56, the box shall show evidence of discoloration when tested as specified in 4.5.5.

3.9.2 Presence of M-GARD W55Ø (zinc naphthenate emulsifiable) preservative. When treated with M-GARD W55Ø, the box shall show evidence of discoloration when treated as specified in 4.5.6.

3.9.3 Presence of M-GARD W51Ø (copper naphthenate) or Cunapsol 5 (copper naphthenate) preservative. When treated with M-GARD W51Ø, the box shall show evidence of discoloration when treated as specified in 4.5.7."

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4.4.3.5 and 4.4.3.6 Delete in their entirety and substitute:

"4.4.3.5 Presence of wood preservative. (See 3.9), Major defect. A sample of 15 boxes shall be selected at random, from each lot, for this test. Four individual boards of each box (cleat, top or bottom board, end board and batten) shall be subjected to the test specified in 4.5.5, 4.5.6, or 4.5.7, as applicable. If one or more boards fail to meet the applicable requirement, the box shall be classed defective. Failure of any box to comply with the requirements shall be cause for rejection of the lot."

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4.5.1 Load test of rope handles. Delete in its entirety and substitute the following paragraphs:

"4.5.1 Load test of rope handles. The box shall be packed with any suitable material until its gross weight is twice as great as its gross weight packed with the item for which it was intended. If the resulting weight of the packed box is 250 pounds (lbs.) or less, 4.5.1.1 shall be followed. If the resulting weight is greater than 250 lbs. and not greater than 500 lbs., 4.5.1.2 shall be followed.

4.5.1.1 Rope handle load test (250 lbs. or less). The box shall be suspended freely for one minute, minimum, from each rope handle in turn. In the event that one or both handles fail the requirement, the box shall be classed defective.

4.5.1.2 Rope handle load test (from greater than 250 lbs. to 500 lbs., maximum). The box shall suspend freely, for one minute, minimum, from both rope handles (at once). In the event that one or both handles fail the requirement, the box shall be classed defective."

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4.5.6 Presence of M-GARD W550 (zinc naphthenate emulsifiable) preservative: Delete in its entirety and substitute:

"4.5.6 Presence of M-GARD W550 (zinc naphthenate emulsifiable) preservative.

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4.5.6.1 Materials and equipment. The material and equipment required are as follows:

a. Reagent. Dissolve 0.1 grams of dithizone (diphenylthiocarbazone) (see 6.12) in either 100 ml of methyl ethyl ketone or 100 ml of chloroform. (Note: This solution may be stored for long periods of time.)

b. Eye dropper. An ordinary glass tube eyedropper may be used.

4.5.6.2 Test procedure. Five drops of the solution shall be applied to the wood surface. The indicator will turn red when zinc (M-GARD W550) is present. The color green indicates that the preservative is not present. Both colors face quickly in the presence of light."

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After 4.5.6.2.2 Test procedure. Add new paragraphs as follows:

"4.5.7 Presence of M-GARD- W510 (copper naphthenate emulsifiable) or Cunapsol 5 (copper naphthenate) preservative. The materials and equipment required are as follows:

a. Reagent. Dissolve 0.5 grams chrome azurol "S" concentrate (see 6.11) and 5.0 grams sodium acetate in 80 mil of distilled water and then dilute further to 500 ml total with distilled water.

b. Sprayer. A common manual (fly) sprayer type applicator shall be used.

4.5.7.1 Test procedure. Spray solution over surface of dried treated wood. A deep blue color reveals the presence of copper (from copper naphthenate)."

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6.12 Change to read as follows:

"6.12 Dithizone. Dithizone (diphenylthiocarbazone) may be obtained from J. T. Baker Chemical Co., Phillipsburg, New Jersey 08865 or equivalent facility."

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Add new paragraphs as follows:

"6.14 M-GARD W510. M-GARD W510 (copper naphthenate), for 2 percent copper as metal, may be obtained from Mooney Chemicals Inc., 2301 Scranton Road, Cleveland, Ohio 44113-9988 or equivalent facility.

6.15 Cunapsol 5. Cunapsol 5 (copper naphthenate), for 2 percent copper as metal, may be obtained from Chapman Chemical Company, P.O. Box 9158, Memphis, TN 38019 or equivalent facility."

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